Some **non-equivalences** (inequivalence)

$$\exists x\, (P(x) \wedge Q(x)) \quad \not\equiv \quad \exists x\, P(x) \wedge \exists x\, Q(x),$$
$$\forall x\, (P(x) \vee Q(x)) \quad \not\equiv \quad \forall x\, P(x) \vee \forall x\, Q(x).$$

Hint: It is enough to think of x in $\{1, 2\}$.

4 propositions in total: P(1), P(2), Q(1) and Q(2).
We find one case where LHS and RHS have different truth values.
By trial-and-error, we find: P(1) =T, P(2) = F, Q(1) = F, Q(2) = T.

LHS = $\exists$ x (P(x) $\wedge$ Q(x)) = (P(1) $\wedge$ Q(1)) $\vee$ (P(2) $\wedge$ Q(2))
= (T and F) or (F and T) = F or F = F.
RHS = $\exists$ x P(x) $\wedge$ $\exists$ x Q(x) = (P(1) $\vee$ P(2)) $\wedge$ (Q(1) $\vee$ Q(2))
= (T or F) and (F or T) = T and T = T.

In this case, LHS and RHS have different truth values,
so LHS is not equal to the RHS.

De Morgan's laws for quantifiers.

$$\neg (\forall x\ P(x)) \equiv \exists x\ \neg P(x),$$
$$\neg (\exists x\ P(x)) \equiv \forall x\ \neg P(x).$$

Again, think of the finite domain case: D = {1,2}
Two propositions: P(1), P(2)

LHS = not (forall x, P(x)) = not(P(1) and P(2))

= not P(1) or not P(2) (De Morgan's law)

= thereexists x , not P(x)
= RHS.

Remark. It applies to the infinite domain case too.

1.5 Nested Quantifiers

Example.
x in $\mathcal{D}_1 = \{$students in University A$\}$,
y in $\mathcal{D}_2 = \{$students in University B$\}$, where $B \neq A$,

$$F(x, y) = (\text{x and y are friends}).$$

- "$\forall x \, \forall y \;\; F(x, y)$" means

Forall x in Univ A   and   for all y in Univ B,   x and y are friends.

This means: Any student in Univ A is a friend to all/every students in B.

The proposition "forall x forall y, F (x, y)" is (almost 100%) F..

1.5 Nested Quantifiers

Example.
x in $\mathcal{D}_1 = \{$students in University A$\}$,
y in $\mathcal{D}_2 = \{$students in University B$\}$, where $B \neq A$,

$$F(x, y) = (\text{x and y are friends}).$$

- "$\forall x \, \forall y \quad F(x, y)$" means

- "$\exists x \, \exists y \quad F(x, y)$" means

Thereexists x in Univ A and thereexists y in Univ B,
x and y are friends.

This means: We can find (at least) one student x in Univ A, and
find another in Univ B, such that x and y are friends.

The proposition "thereexists x thereexists y, F (x, y)" is (very likely) T.

Example.

x in $\mathcal{D}_1 = \{$students in University A$\}$,
y in $\mathcal{D}_2 = \{$students in University B$\}$, where $B \neq A$,

$$F(x, y) = (\text{x and y are friends}).$$

- "$\forall x\, \forall y \;\; F(x, y)$" means

- "$\exists x\, \exists y \;\; F(x, y)$" means

Example

- "$\forall x, \forall y$ in $\mathcal{D} = \{$real numbers$\}$, $(x + y = y + x)$"

commutative law for addition.

The Order of Quantifiers

$$\begin{aligned}\forall x\,\forall y, P(x,y) &\equiv \forall y\,\forall x, P(x,y),\\ \exists x\,\exists y, P(x,y) &\equiv \exists y\,\exists x, P(x,y),\end{aligned}$$

but

$$\forall x, \exists y, P(x,y) \not\equiv \exists y, \forall x, P(x,y).$$

Example. x and y are two students in XMUM, and $P(x,y)$: (x and y have a common course).

- $\forall x, \exists y, P(x,y)$ means

For any student x in XMUM, they can find another student y (a classmate fo x) such that they have a common course.

“ Forall x, thereexists y , P(x, y)” is T

The Order of Quantifiers

$$\begin{aligned} \forall x \, \forall y, P(x,y) &\equiv \forall y \, \forall x, P(x,y), \\ \exists x \, \exists y, P(x,y) &\equiv \exists y \, \exists x, P(x,y), \end{aligned}$$

but

$$\forall x, \exists y, P(x,y) \not\equiv \exists y, \forall x, P(x,y).$$

Example. x and y are two students in XMUM, and $P(x,y)$: (x and y have a common course).

- $\forall x, \exists y, P(x,y)$ means
- $\exists x, \forall y, P(x,y)$ means

There is one (very special, maybe too special) student in XMUM, where he/she has a common course to every student in XMUM.

“ Thereexists x, forall y , P(x, y)” is F.

Remember

When $\forall$ and $\exists$ are nested together, the order is important.

Example

- $\forall x \, \exists y \quad (x + y = 0)$

For any number x (given), we can find
another number y (depending on the given number x) such that x+ y =0.
We can write y in terms of x, in this case y = —x.

- $\exists x \, \forall y \quad (x + y = 0)$

Thereexists a number x (fixed), such that x + y = 0 for any value y (random).
No such number x exists, so the statement if F.

Example: ∃x ∀y (xy = 0)
Find/solve for one x such that (xy =0 for any choice of y) .
This time x exists, which is 0.